DSA

Practical -02

Hashing

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Batch:A2

Hashtable code:-

```
#include <iostream>
using namespace std;

const int SIZE = 10;
int hashTable[SIZE];
bool occupied[SIZE];

int hashFunction(int key) {
    return key % SIZE;
}

void showmore() {
    cout << "\nSelect your choice:" << endl;
    cout << "1. Put the value" << endl;
    cout << "2. Remove the value" << endl;
    cout << "3. Search the value" << endl;
    cout << "4. Display values" << endl;
    cout << "5. Exit" << endl;
}

void insert() {
```

```
    int key;
    cout << "Enter the number to insert: ";
    cin >> key;
    int index = hashFunction(key);
    int originalIndex = index;

    bool inserted = false;
    for (int i = 0; i < SIZE; ++i) {
        if (!occupied[index]) {
            hashTable[index] = key;
            occupied[index] = true;
            cout << "Inserted " << key << " at index " << index <<
endl;
            inserted = true;
            break;
        }
        index = (index + 1) % SIZE;
    }

    if (!inserted) {
        cout << "Hash table is full. Cannot insert " << key << endl;
    }
```

```
}

void removeValue() {
    int key;
    cout << "Enter the number to remove: ";
    cin >> key;
    int index = hashFunction(key);

    for (int i = 0; i < SIZE; ++i) {
        if (occupied[index] && hashTable[index] == key) {
            occupied[index] = false;
            cout << "Removed " << key << " from index " << index
<< endl;
            return;
        }
        index = (index + 1) % SIZE;
    }

    cout << "Value not found.\n";
}

void search(int key) {
```

```
    int index = hashFunction(key);

    for (int i = 0; i < SIZE; ++i) {
        if (occupied[index] && hashTable[index] == key) {
            cout << "Found " << key << " at index " << index <<
endl;
            return;
        }
        index = (index + 1) % SIZE;
    }

    cout << "Value not found.\n";
}

void display() {
    cout << "\nHash Table Contents:\n";
    for (int i = 0; i < SIZE; ++i) {
        cout << "Bucket " << i << ": ";
        if (occupied[i])
            cout << hashTable[i];
        else
            cout << "Empty";
```

```
        cout << endl;
    }
}

int main() {
    int choice, value;

    // Initialize table
    for (int i = 0; i < SIZE; ++i) {
        occupied[i] = false;
    }

    do {
        showmore();
        cout << "Enter your choice: ";
        cin >> choice;

        switch (choice) {
            case 1:
                insert();
                break;
            case 2:
```

```
            removeValue();
            break;
        case 3:
            cout << "Enter the value to search: ";
            cin >> value;
            search(value);
            break;
        case 4:
            display();
            break;
        case 5:
            cout << "Exiting program.\n";
            break;
        default:
            cout << "Wrong choice. Please try again.\n";
      }
   } while (choice != 5);

   return 0;
}
```

Output:-

/*

Select your choice:

1. Put the value

2. Remove the value

3. Search the value

4. Display values

5. Exit

Enter your choice: 1

Enter the number to insert: 10

Inserted 10 at index 0

Select your choice:

1. Put the value

2. Remove the value

3. Search the value

4. Display values

5. Exit

Enter your choice: 1

Enter the number to insert: 11

Inserted 11 at index 1

Select your choice:

1. Put the value

2. Remove the value

3. Search the value

4. Display values

5. Exit

Enter your choice: 1

Enter the number to insert: 12

Inserted 12 at index 2

Select your choice:

1. Put the value

2. Remove the value

3. Search the value

4. Display values

5. Exit

Enter your choice: 1

Enter the number to insert: 21

Inserted 21 at index 3

Select your choice:

1. Put the value
2. Remove the value
3. Search the value
4. Display values
5. Exit

Enter your choice: 2

Enter the number to remove: 12

Removed 12 from index 2

Select your choice:

1. Put the value
2. Remove the value
3. Search the value
4. Display values
5. Exit

Enter your choice: 3

Enter the value to search: 11

Found 11 at index 1

Select your choice:

1. Put the value
2. Remove the value

3. Search the value

4. Display values

5. Exit

Enter your choice: 4

Hash Table Contents:

Bucket 0: 10

Bucket 1: 11

Bucket 2: Empty

Bucket 3: 21

Bucket 4: Empty

Bucket 5: Empty

Bucket 6: Empty

Bucket 7: Empty

Bucket 8: Empty

Bucket 9: Empty

Select your choice:

1. Put the value

2. Remove the value

3. Search the value

4. Display values

5. Exit

Enter your choice: 5

Exiting program.

*/

Car Parking Code:-

```
#include <iostream>
using namespace std;

int slots[10];
bool taken[10];

int getSlot(int regNo) {
    return regNo % 10;
}

void parkCar() {
    int regNo;
    cout << "Enter car number: ";
    cin >> regNo;
```

```
    int pos = getSlot(regNo);

    for (int i = 0; i < 10; i++) {
        if (!taken[pos]) {
            slots[pos] = regNo;
            taken[pos] = true;
            cout << "Car parked at slot " << pos << endl;
            return;
        }
        pos = (pos + 1) % 10;
    }

    cout << "Parking full!\n";
}

void removeCar() {
    int regNo;
    cout << "Enter car number to remove: ";
    cin >> regNo;

    int pos = getSlot(regNo);
```

```
    for (int i = 0; i < 10; i++) {
        if (taken[pos] && slots[pos] == regNo) {
            taken[pos] = false;
            cout << "Car removed from slot " << pos << endl;
            return;
        }
        pos = (pos + 1) % 10;
    }

    cout << "Car not found!\n";
}

void showParking() {
    cout << "\nParking Slots:\n";
    for (int i = 0; i < 10; i++) {
        cout << "Slot " << i << ": ";
        if (taken[i])
            cout << slots[i] << endl;
        else
            cout << "Empty\n";
    }
}
```

```
int main() {
    int ch;

    for (int i = 0; i < 10; i++)
        taken[i] = false;

    do {
        cout << "\n1. Park Car\n2. Remove Car\n3. Show
Parking\n4. Exit\n";
        cout << "Enter choice: ";
        cin >> ch;

        if (ch == 1)
            parkCar();
        else if (ch == 2)
            removeCar();
        else if (ch == 3)
            showParking();
        else if (ch == 4)
            cout << "Bye!\n";
        else
```

```
            cout << "Invalid choice\n";

    } while (ch != 4);

    return 0;
}
```

Output:-

```
/*

1. Park Car
2. Remove Car
3. Show Parking
4. Exit
Enter choice: 1
Enter car number: 4568
Car parked at slot 8

1. Park Car
2. Remove Car
3. Show Parking
```

4. Exit

Enter choice: 1

Enter car number: 7894

Car parked at slot 4

1. Park Car

2. Remove Car

3. Show Parking

4. Exit

Enter choice: 1

Enter car number: 7412

Car parked at slot 2

1. Park Car

2. Remove Car

3. Show Parking

4. Exit

Enter choice: 2

Enter car number to remove: 7894

Car removed from slot 4

1. Park Car

2. Remove Car

3. Show Parking

4. Exit

Enter choice: 3

Parking Slots:

Slot 0: Empty

Slot 1: Empty

Slot 2: 7412

Slot 3: Empty

Slot 4: Empty

Slot 5: Empty

Slot 6: Empty

Slot 7: Empty

Slot 8: 4568

Slot 9: Empty

1. Park Car

2. Remove Car

3. Show Parking

4. Exit

Enter choice: 4

Bye!
*/